Second rating of the XAI comparison table: guide for Imansha

Oct 9, 2026 · @Minidu

Please rate the 24 cells of the XAI comparison table on your own, from Supun's raw outputs only, and fill in the sheet below by Oct 12, 2026 morning. It takes about an hour. Your ratings make the table two independent judgements instead of one, which a reviewer will look for.

What you get, and what not to look at

You rate from the raw outputs only, so the result counts as an independent second rating.

Supun will send you:

- The raw SHAP, LIME and DiCE outputs for one prediction, exactly as the tools produced them (the paper's raw-output figure).
- Nothing else from his table: no ratings, no reasons, no "our adapted explanation" column.

The prediction they explain (synthetic): Hanwella gauge on the Kelani Ganga. The model gives a 0.29 chance (about 30%) of the river reaching the Minor Flood Level on Thursday. Yesterday's forecast was 0.10.

Do not look at, before you finish:

- Supun's ratings, in any version.
- Task 2b of the Research Approach (v3 or v3.1). Its table contains Supun's first-pass ratings.
- Do not discuss the table with Supun or Minindu until the 12 Oct comparison.

If you have already read the Task 2b table, that is fine: tick the box at the top of the rating sheet. We report it honestly rather than claim a fully blind rating.

For the guideline wording, use the XAI guideline evidence catalogue (`docs/research/xai-guideline-evidence-catalogue.pdf`). Read only the guideline name and its "Implication for the explanation" column, not the evidence.

The eight needs

Each row of the table is one thing a resident needs from an explanation, taken from one or two guidelines. A Yes means the raw output, as shown, meets that need for a lay resident.

#	Need	Guidelines	Yes means the output…
1	Bottom line first	G21 Provide summary of long documents	opens with a one-sentence statement of what is predicted
2	Likelihood without numeracy	G26 Alternatives for numerical concepts; G119 CAP message structure	shows how likely it is without needing a decimal or percentage (a word, picture or "3 in 10"), using a standard likelihood word
3	Place and time window	G73 Persistent visual context	says where and when the prediction applies
4	Consequence and action	G50 State results of actions	says what it means for the person and what to do
5	Familiar references	G38 Familiar metrics and units	relates quantities to things a resident knows (a landmark, everyday units, a weekday)
6	What changed	G89 Warning before any change	says how this prediction differs from the previous one
7	Why, in plain words	G25 Explain implied content	gives the reasons in everyday words a resident can follow
8	Hazard named	G11 Most important stands out; G120 Universal hazard symbols	names the hazard, flood, in words or a standard symbol

How to rate

One rating and one reason per cell; judge what a resident would see, not what a developer could build from it.

- **Yes**: the need is met as the output stands.
- **Partly**: some of it is there, but a resident would still have to work something out, or a part is missing.
- **No**: the need is not met.

Rules:

1. Rate the raw output exactly as produced. Do not credit information that could be added later, or that is in the model but not in the output.
2. Read each output as a lay resident with no ML knowledge would.
3. Rate each cell on its own. A method that scores No on one need can still score Yes on another.
4. Write a one-line reason for every cell, naming what in the output decided it (for example, a value, a label or a missing item).
5. If you are unsure between two ratings, pick one and say why in the reason. Do not leave a cell empty.
6. Work alone, and do not change a rating after the 12 Oct comparison starts.

Rating sheet

Pick a rating from each chip and write the reason beside it; this table is what we compare on 12 Oct.

- [ ] I had already seen the Research Approach Task 2b table before rating

#	Need	Method	Rating	Reason (one line)
1	Bottom line first (G21)	SHAP	▾	
2	Bottom line first (G21)	LIME	▾	
3	Bottom line first (G21)	Counterfactual (DiCE)	▾	
4	Likelihood without numeracy (G26, G119)	SHAP	▾	
5	Likelihood without numeracy (G26, G119)	LIME	▾	
6	Likelihood without numeracy (G26, G119)	Counterfactual (DiCE)	▾	
7	Place and time window (G73)	SHAP	▾	
8	Place and time window (G73)	LIME	▾	
9	Place and time window (G73)	Counterfactual (DiCE)	▾	
10	Consequence and action (G50)	SHAP	▾	

#	Need	Method	Rating	Reason (one line)
11	Consequence and action (G50)	LIME	▾	
12	Consequence and action (G50)	Counterfactual (DiCE)	▾	
13	Familiar references (G38)	SHAP	▾	
14	Familiar references (G38)	LIME	▾	
15	Familiar references (G38)	Counterfactual (DiCE)	▾	
16	What changed (G89)	SHAP	▾	
17	What changed (G89)	LIME	▾	
18	What changed (G89)	Counterfactual (DiCE)	▾	
19	Why, in plain words (G25)	SHAP	▾	
20	Why, in plain words (G25)	LIME	▾	
21	Why, in plain words (G25)	Counterfactual (DiCE)	▾	
22	Hazard named (G11, G120)	SHAP	▾	
23	Hazard named (G11, G120)	LIME	▾	
24	Hazard named (G11, G120)	Counterfactual (DiCE)	▾	

After you finish: the 12 Oct comparison

On 12 Oct you and Supun put your two sheets side by side; Minindu joins to record the result.

1. Count the cells where both ratings match exactly (N of 24). This raw agreement is what the paper reports first.
2. Also compute weighted kappa (ordered: No < Partly < Yes), as your literature review recommends for ordinal ratings. With 24 cells it is a supporting number, not the headline.
3. For each cell that differs, both read your reasons aloud, then agree a final rating and a one-line final reason. Keep both original ratings.

4. Note whether either rater had seen the Task 2b table (the tick box above).

The paper will say: "Two authors rated the 24 cells independently and agreed on N of 24 before discussion."